



Scoring Methodology

Ratios, weights, thresholds and scenario analysis

This document sets out in full how a company's readiness band comes about: which ratios feed it, where every single threshold comes from, and what is deliberately not modelled. It is written so that a tax adviser or a credit analyst can recompute every value and disagree with every assumption.

13

weighted factors

6

empirically calibrated

46%

calibrated weight

3

stress scenarios

8

knock-out criteria

SECTION 1

01 Purpose and scope

The method produces an **indicative readiness band** (A to E). It describes how a credit application with this data behind it is likely to read to a typical lender – and which parts of that are changeable.

What this explicitly is not. Not a rating within the meaning of EU Regulation (EC) No 1060/2009, not a probability of default, not a credit decision and not an offer. No PD is estimated and none is published. Every credit decision is taken solely by the lender concerned, on its own standards.

Three construction rules follow from that, and they are enforced in code, not merely asserted here:

- **Traceability.** Every factor exposes its raw value, its breakpoints, its weight and the points it contributed. There is no figure that cannot be explained line by line.
- **No machine learning.** Piecewise-linear interpolation over published thresholds, and nothing else.
- **No rating vocabulary.** The words are „band“ and „indicative“.

SECTION 2

02 Regulatory basis

The choice of ratios does not come from a textbook. It comes from three sources that describe what lenders in the euro area are required to examine, and what they in fact examine:

Source	What is taken from it
EBA/GL/2020/06 Guidelines on loan origination and monitoring, in force since 30 June 2021	Annex 3 section B names the metrics an institution should consider for lending to enterprises (section 3). Paragraphs 121 and 128 govern what belongs in the analysis of the financial position; paragraph 120 makes clear that collateral must <i>not</i> be a predominant criterion; paragraphs 131 and 156–158 require the scenario analysis.

Source	What is taken from it
MaRisk (BaFin) Circular 10/2021, Annex 1, BTO 1.2	BTO 1.2.1 para. 1 requires the risk factors to be analysed „with particular regard to debt service capacity“ – which is why debt service coverage carries the largest single weight here. BTO 1.4 para. 3 requires qualitative criteria alongside quantitative ones.
Bundesbank credit assessment system (ICAS) Description of the procedure, as at December 2023	The ratios with which a central bank assesses the creditworthiness of German companies from HGB accounts: EBITDA, adjusted total debt, liquidity, return on sales, debt repayment capability, adjusted equity ratio and days payable. Also the empirical distributions (section 6) and the method behind the scenario analysis (section 8).

The most important finding from reading these sources: none of them sets threshold values. EBA Annex 1 requires the institution to *define* „acceptable ... ratio limits“, but names no figures. MaRisk governs the process, not the level of any ratio. The supervisor therefore settles *which* ratios and *that* limits must exist – how high is a separate decision that has to be justified on its own. That is precisely why this document presents and evidences the choice of ratios (section 3) and the thresholds (sections 5 and 6) separately.

03 Mapped against EBA Annex 3

Annex 3 section B of the EBA guidelines lists twenty metrics for lending to enterprises. The guideline requires them to be considered „to an extent that is applicable and appropriate to the specific credit proposal“ (para. 128 e). The table below states, for every one of them, what happens – including the cases where deliberately nothing does.

No.	Metric per EBA Annex 3 B	Covered	How it is handled
6	Equity ratio	yes	Equity ratio (economic), weight 0.16
7	(Long-term) debt-to-equity ratio	partly	Covered through the equity ratio and dynamic gearing; not weighted separately, so that leverage is not counted three times.
8	EBITDA	yes	Input to the DSCR, to gearing and to the margin
9	Debt yield	no	A property metric, for real estate finance
10	Interest bearing debt / EBITDA	yes	Dynamic gearing, weight 0.11
11	Enterprise value	no	Requires market values; not available for private SMEs
12	Capitalisation rate	no	A property metric, for real estate finance
13	Asset quality	partly	Through days receivable, days payable and fixed-asset coverage; no metric of its own
14	Total debt service coverage ratio	yes	Debt service coverage, weight 0.20
15	Cash debt coverage ratio	no	Requires a cash flow statement; small companies do not prepare one
16	Coverage ratio (current assets / short-term debt)	yes	Quick ratio, weight 0.07 (stricter than the current ratio)
17	Future cash flow analysis	partly	Through the scenario analysis, not through a forecast
18	Return on assets	yes	Return on assets, weight 0.05
19	Debt service	yes	The denominator of debt service coverage
20	Loan to cost	no	A project finance metric
21	Interest coverage ratio	yes	Interest coverage, weight 0.05
22	Return on equity ratio	no	Unstable to the point of being misleading where book equity is thin or negative - which is common in owner-managed SMEs
23	Return on capital employed	partly	Through return on assets
24	Net profit margin	yes	EBIT margin, weight 0.09
25	Turnover evolution	partly	Computed and shown in the report, but not weighted: a single year-on-year comparison does not carry a weight

04 Factors and weights

The weights sum to 100%. Where a factor cannot be computed for want of data, the remaining weights are scaled up proportionally and the report states the coverage achieved. A thin file is not punished for being thin – it is labelled as thin.

Factor	Weight	Type	Regulatory origin
Equity ratio (economic) eigenkapitalquote	16%	calibrated	EBA Annex 3 no. 6; Bundesbank model ratio
Debt service coverage incl. new facility (DSCR) kapitaldienstfaehigkeit_inkl_neu	20%	convention	EBA Annex 3 no. 14, 19; MaRisk BTO 1.2.1 para. 1
Dynamic gearing (net debt / EBITDA) dynamischer_verschuldungsgrad	11%	convention	EBA Annex 3 no. 10; Bundesbank additional ratio
EBIT margin ebit_marge	9%	calibrated	EBA Annex 3 no. 24; Bundesbank model ratio
Quick ratio liquiditaet_2_grades	7%	calibrated	EBA Annex 3 no. 16; Bundesbank model ratio
Interest coverage (EBIT / interest) zinsdeckungsgrad	5%	convention	EBA Annex 3 no. 21
Return on assets (net result + interest) gesamtkapitalrentabilitaet_bbk	5%	calibrated	EBA Annex 3 no. 18
Fixed-asset coverage II anlagendeckungsgrad_ii	5%	calibrated	Golden balance-sheet rule; Bundesbank ratio
Days payable outstanding kreditorenlaufzeit_tage	4%	calibrated	Bundesbank additional ratio (days payable)
Overdraft utilisation kontokorrent_auslastung	5%	convention	EBA para. 121 d (drawings on committed lines)
Currency of the monthly accounts (BWA) bwa_age_months	4%	convention	EBA Annex 2 B no. 3 (currency of the accounts)
Creditreform credit index creditreform_bonitaetsindex	5%	convention	EBA Annex 2 B no. 8, 9 (external report/rating)
Payment behaviour zahlungsverhalten	4%	convention	EBA para. 121 d (payment behaviour, tax and social security)

One exception to the redistribution. Where EBITDA is zero or negative, dynamic gearing is arithmetically undefined. The factor is then not dropped but scored at zero: a company with no operating surplus cannot repay its debt out of the business, and dropping the factor would flatter exactly the case that matters.

Why collateral is not scored. EBA para. 120: collateral is the second way out, not the primary source of repayment, and cannot by itself justify a lending decision. A collateral shortfall therefore appears as a finding and an action in the report, but not as a factor that moves points.

SECTION 5

05 The thresholds in full

Between two breakpoints the score is interpolated linearly; beyond them it is clamped. Every point value can therefore be recomputed by hand – that is the reason for this construction, not a side effect of it.

Factor	Breakpoints (ratio value = points, linearly interpolated)
Equity ratio (economic) Bundesbank: Equity / total assets	$-20,0\% = 0 \rightarrow 0,0\% = 25 \rightarrow 5,0\% = 38 \rightarrow 13,0\% = 58 \rightarrow 33,2\% = 70 \rightarrow 56,6\% = 82 \rightarrow 80,0\% = 95 \rightarrow 100,0\% = 100$
Debt service coverage incl. new facility (DSCR) Convention	$0,70x = 0 \rightarrow 0,90x = 12 \rightarrow 1,00x = 25 \rightarrow 1,10x = 42 \rightarrow 1,20x = 58 \rightarrow 1,30x = 72 \rightarrow 1,50x = 88 \rightarrow 2,00x = 100$
Dynamic gearing (net debt / EBITDA) Convention	$0,00x = 100 \rightarrow 1,00x = 95 \rightarrow 2,00x = 88 \rightarrow 3,00x = 75 \rightarrow 3,50x = 65 \rightarrow 4,00x = 55 \rightarrow 5,00x = 38 \rightarrow 6,00x = 25 \rightarrow 8,00x = 10 \rightarrow 12,00x = 0$
EBIT margin Bundesbank: Pre-tax result / revenue	$-10,0\% = 0 \rightarrow -5,0\% = 12 \rightarrow -2,0\% = 28 \rightarrow 0,0\% = 45 \rightarrow 1,4\% = 58 \rightarrow 4,5\% = 70 \rightarrow 9,7\% = 82 \rightarrow 18,0\% = 95 \rightarrow 25,0\% = 100$
Quick ratio Bundesbank: Quick ratio	$0,0\% = 0 \rightarrow 20,0\% = 25 \rightarrow 30,0\% = 40 \rightarrow 53,1\% = 58 \rightarrow 115,3\% = 70 \rightarrow 254,3\% = 82 \rightarrow 400,0\% = 95 \rightarrow 600,0\% = 100$
Interest coverage (EBIT / interest) Convention	$0,00x = 0 \rightarrow 1,00x = 20 \rightarrow 1,50x = 35 \rightarrow 2,00x = 50 \rightarrow 3,00x = 70 \rightarrow 5,00x = 88 \rightarrow 8,00x = 100$
Return on assets (net result + interest) Bundesbank: (Net result + interest) / total assets	$-5,0\% = 0 \rightarrow 0,0\% = 30 \rightarrow 2,3\% = 58 \rightarrow 6,7\% = 70 \rightarrow 14,1\% = 82 \rightarrow 22,0\% = 95 \rightarrow 30,0\% = 100$
Fixed-asset coverage II Bundesbank: Long-term capital / fixed assets	$50,0\% = 0 \rightarrow 80,0\% = 12 \rightarrow 100,0\% = 30 \rightarrow 101,2\% = 58 \rightarrow 195,2\% = 70 \rightarrow 533,7\% = 82 \rightarrow 800,0\% = 95 \rightarrow 1.200,0\% = 100$
Days payable outstanding Bundesbank: Trade payables / materials expense	$0,0 = 82 \rightarrow 15,3 = 82 \rightarrow 31,6 = 70 \rightarrow 59,5 = 58 \rightarrow 75,0 = 38 \rightarrow 100,0 = 20 \rightarrow 120,0 = 0$
Overdraft utilisation Convention	$0,0\% = 100 \rightarrow 30,0\% = 92 \rightarrow 50,0\% = 80 \rightarrow 70,0\% = 62 \rightarrow 80,0\% = 45 \rightarrow 90,0\% = 28 \rightarrow 100,0\% = 10 \rightarrow 110,0\% = 0$
Currency of the monthly accounts (BWA) Convention	$1,0 = 100 \rightarrow 2,0 = 90 \rightarrow 3,0 = 72 \rightarrow 4,0 = 55 \rightarrow 6,0 = 32 \rightarrow 9,0 = 12 \rightarrow 12,0 = 0$
Creditreform credit index Convention	$100 = 100 \rightarrow 150 = 95 \rightarrow 200 = 85 \rightarrow 250 = 72 \rightarrow 300 = 55 \rightarrow 350 = 35 \rightarrow 400 = 18 \rightarrow 500 = 5 \rightarrow 600 = 0$
Payment behaviour Convention	$0 = 0 \rightarrow 100 = 100$

06 Where the thresholds come from

Calibrated factors

Six of the 13 factors – together 46% of the weight – are tied to the **quartile values of the Deutsche Bundesbank's annual accounts statistics** (Verhaeltniszahlen): all sectors, the EUR 2–10m and 10–50m revenue size classes averaged, latest reporting year. Data as at: Deutsche Bundesbank, Jahresabschlussstatistik (Verhaeltniszahlen), Ausgabe 2026, Berichtsjahr 2023, Quartilswerte.

The anchoring is:

25th percentile

= 58 points (lower edge of band C)

Median

= 70 points (middle of band B)

75th percentile

= 82 points (band A)

The reasoning is one step long: the ECB's SAFE survey puts roughly 14% of applicants in significant financing difficulty. A company at the median of its size class is therefore not a problem case but financeable – band B. Mapping percentiles directly would put the median firm at 50 points and so assert that half of the German Mittelstand is a borderline case. That would be empirically wrong.

For „lower is better“ ratios the anchors invert. For days payable the 25th percentile is the good end, so the mapping there is 25th percentile = 82, median = 70, 75th percentile = 58. That curve is also the only one capped below 100: paying suppliers quickly is not evidence of creditworthiness, only the absence of a warning sign.

The ends beyond the quartiles. The statistics give three points, not a distribution. The ends of each curve are therefore not extrapolated but drawn to economically meaningful limits – zero equity, a negative margin, fixed-asset coverage below 100%.

Factors that are not calibrated

For the remaining factors the publication carries no comparable series. Their curves are reasoned convention and are labelled as such: debt service coverage, dynamic gearing, interest coverage, overdraft utilisation, currency of the monthly accounts, the Creditreform index and payment behaviour.

One translation deliberately not made. The Bundesbank publishes „cash flow as a percentage of net borrowed funds“. That is *not* our dynamic gearing: a different numerator (cash flow after interest and tax, rather than EBITDA) and a far wider denominator (all liabilities, rather than financial debt alone). Converting one into the other would be a guess dressed up as a calibration. The factor therefore stays convention – and is the first candidate for revision once the outcome log holds enough real placements.

SECTION 7

07 From points to bands

Band	Points	Meaning
A	78,0 – 100,0	Likely financeable on standard terms
B	65,0 – 78,0	Financeable; terms probably improvable
C	52,0 – 65,0	Borderline - the typical decline and mispricing zone
D	38,0 – 52,0	Decline likely without restructuring or a guarantee
E	0,0 – 38,0	Substantial credit weakness

The band thresholds themselves are not empirically derived. They are set so that the median firm lands in band B and the lower quartile in band C – that is, from the same SAFE reasoning as the factor anchors, not from observed decline rates. Anyone attacking this mapping is attacking the weakest point of the method, and rightly so. It changes once enough documented credit decisions exist.

SECTION 8

08 Scenario analysis

A point-in-time ratio from last year's accounts answers the wrong question. The lender does not want to know how the company looked on 31 December, but whether it still carries the debt service when things go against it. Both supervisory sources require this explicitly (EBA paras. 131 and 156–158, MaRisk BTO 1.2.1 para. 1).

Scenario	Assumption and source
Interest rates +200 basis points	All credit facilities become 2.00 percentage points more expensive (EBA/GL/2020/06 para. 158 k). For comparison: in the adverse scenario of the 2025 EU-wide stress test German long-term rates rise by about 1.1 percentage points - so this is the harder test.
Revenue down 10%	Revenue -10%, of which 65% variable costs fall away with it; fixed costs remain (EBA/GL/2020/06 para. 158 a). The order of magnitude follows the adverse scenario of the 2025 EU-wide stress test, which assumes German GDP 7.5% lower on a cumulative basis.
Combined: revenue -10% and rates +200 bp	Both events at once (EBA/GL/2020/06 para. 156 expressly permits a multifactor analysis).

How the shock is worked through the accounts

The mechanism follows the Deutsche Bundesbank's own published approach to stressing a financial statement inside its credit assessment system (Technical Paper 02/2023, section 5.2). There a cost line is stressed; the plumbing is the same whatever the shock is:

- determine the additional cost, or the contribution lost;
- fund it from cash first, then through short-term bank borrowing;
- raise the interest charge on that new borrowing at the existing average rate, because the interest burden is itself rating-relevant;
- reduce the tax charge at the company's own effective rate (never below zero);
- carry the reduced result through to equity;
- recompute every ratio on the stressed accounts.

The rate shock hits the loan agreements, not just the P&L. Paragraph 158 k speaks of an increase „on all credit facilities of the borrower“. Had only the interest line been raised, debt service – and with it debt service coverage, the decisive ratio – would have stayed unchanged. The rate on each individual facility is therefore raised.

What the scenario analysis is not. Not a forecast and not a statement of probability. It says: on these stated assumptions the band moves from X to Y. The assumptions are printed next to the result so that the owner can disagree with them – they know their own variable cost share better than we do.

SECTION 9

09 Knock-out criteria

A weighted score answers the question of how a file reads overall. It cannot answer whether there is something in it that ends the conversation regardless of the total – a weighted average by construction lets a strong factor offset a fatal one. Those are two different questions and they need two different methods.

These 8 criteria are therefore checked separately and reported separately, ahead of the ratios. None of the thresholds is ours:

Code	Criterion	Condition	Basis
KO01	Loss of half the share capital	Balance-sheet equity < 50% of subscribed capital	Sec. 49(3) GmbHG; Banque de France note 4-/6
KO02	Balance-sheet over-indebtedness	Balance-sheet equity negative	Sec. 19 InsO; EBA/GL/2020/06 para. 128 a
KO03	Overdraft limit breached	Drawings > the advised limit	Article 178 CRR (definition of default)
KO04	Tax arrears	Arrears owed to the tax office	EBA/GL/2020/06 para. 121 d
KO05	Returned direct debits	Returned direct debits in the last 12 months; 3 or more is serious	Banque de France note 8
KO06	Consecutive losses	Net loss in the current and the prior year	Banque de France note 6
KO07	Interest absorbing EBITDA	Interest expense >= 50% of EBITDA; 100% or more is serious	Banque de France note 6
KO08	Accounts too old	Most recent accounts >= 23 months old	Banque de France note X

This check is not a legal assessment. Whether a finding gives rise to an obligation – under Sec. 49(3) GmbHG or Sec. 15a InsO, say – turns on a going-concern prognosis that this method cannot make and does not claim to make. The report names the condition and refers the judgement to the tax adviser or a lawyer.

Why a clean result is a statement in its own right. „None of these criteria applies“ is not the same as a good score. It means the conversation about terms can be had at all – and that is exactly the information a business owner needs before applying.

SECTION 10

10 A worked example

A complete run on the fictional sample case Mueller Praezisionstechnik GmbH. Every value is produced by the running system, not transcribed.

Factor	Value	Points	Weight	Contribution
Equity ratio (economic)	10,1%	49,4	16%	7,91
Debt service coverage incl. new facility (DSCR)	1,50x	88,0	20%	17,60
Dynamic gearing (net debt / EBITDA)	3,34x	68,1	11%	7,49
EBIT margin	4,6%	69,4	9%	6,25
Quick ratio	99,3%	69,8	7%	4,89
Interest coverage (EBIT / interest)	4,00x	79,0	5%	3,95
Return on assets (net result + interest)	7,4%	70,9	5%	3,55
Fixed-asset coverage II	97,0%	27,3	5%	1,36
Days payable outstanding	59,4	52,6	4%	2,11
Overdraft utilisation	84,0%	38,2	5%	1,91
Currency of the monthly accounts (BWA)	5,0	43,5	4%	1,74
Creditreform credit index	248	72,5	5%	3,63
Payment behaviour	70	70,0	4%	2,80
Total				65,2 B

The scenario analysis for the same case:

Scenario	Points	Band	Delta
Base case	65,2	B	–
Interest rates +200 basis points	62,5	C	-2,7
Revenue down 10%	35,7	E	-29,5
Combined: revenue -10% and rates +200 bp	34,3	E	-30,9

How to read it: the business is financeable at the balance sheet date (band B), but does not survive a 10% fall in revenue, because debt service coverage then drops below 1.0. That is information which surfaces in the credit conversation anyway – only without preparation.

SECTION 11

11 Limitations of the method

A complete list of what this method cannot do. It belongs in this document, because a description of a method without a list of its weaknesses serves as a sales brochure and as nothing else.

- **No default validation.** The weights and band thresholds are anchored in distributions and in professional judgement, not in observed defaults. There is as yet no outcome data against which the method could be back-tested.
- **Vintage of the benchmark data.** Deutsche Bundesbank, Jahresabschlussstatistik (Verhaeltniszahlen), Ausgabe 2026, Berichtsjahr 2023, Quartilswerte. Erfassungsgrad der Umsatzklasse 2-10 Mio EUR rund 14%, 10-50 Mio EUR rund 42% -- kleinere Unternehmen sind im Datenpool der Bundesbank unterrepraesentiert.
- **Qualitative factors are largely absent.** MaRisk BTO 1.4 para. 3 requires qualitative criteria. Only the currency of the accounts, the external credit report and payment behaviour are covered. Management quality, market position, customer and supplier concentration (EBA paras. 132–136) do not enter the score; they appear as findings.
- **No ESG assessment.** EBA paras. 126–127 require ESG risks to be assessed. The method does not currently do this.

- **No forecast.** EBA para. 129 expects financial projections. What is assessed is the most recent accounts plus scenarios; a missing integrated forecast is reported as a finding, not substituted for.
- **Sector differentiation only in the comparison.** The thresholds apply across sectors; only the sector comparison in the report is sector-specific. The Bundesbank, by contrast, reselects its model ratios for each sector.
- **One factor can drop out.** Service businesses with no materials expense have no days-payable figure; its weight is then redistributed and coverage falls accordingly.

SECTION 12

12 Changes in this version

The result of reading the EBA guidelines, the MaRisk explanatory notes and the Bundesbank publications:

- **Three factors added**, all three empirically calibrated: return on assets (EBA Annex 3 no. 18), fixed-asset coverage II and days payable (both Bundesbank ratios). The calibrated share of the weight rises from 40% to 46%.
- **Weights rebalanced** so that the total stays at 100%. Debt service coverage stays at 20% and remains the largest single weight – MaRisk BTO 1.2.1 para. 1.
- **Scenario analysis introduced** (section 8). This was the largest gap against the guidelines: required by paras. 131 and 156–158, and previously not implemented at all.
- **Return on assets on the Bundesbank definition** (net result plus interest) rather than on an EBIT basis, so that the published quartiles apply without conversion. Both variants are computed; the adopted one is scored.
- **Two previously unused Bundesbank series** added to the dataset the application loads at runtime.
- **Knock-out criteria introduced** (section 9). From reading the EBA benchmarking reports, the Banque de France rating guide and the definition of default in Article 178 CRR. The score alone cannot express a knock-out, because a weighted average offsets it.
- **Overdraft breach upgraded.** Previously a mere warning in the plausibility check. Under Article 178 CRR an overdraft counts as past due once the advised limit is exceeded – that is an indicator of default, not an anomaly.
- **Scenario sizes anchored to the supervisor.** The 10% fall in revenue was previously simply set; it now stands next to the adverse scenario of the 2025 EU-wide stress test, which assumes German GDP 7.5% lower on a cumulative basis. The same comparison shows, conversely, that our 200 basis point rate shock is nearly twice the roughly 110 basis points assumed there.

This document is generated, not maintained. Every factor, weight, breakpoint, band threshold, scenario and the worked example are read from the production modules on each run (`scripts/build_methodology_pdf.py`). A divergence between this paper and the application is not possible.